

GitHub Repository and Completed Files

Project artefacts include:

- API and web-scraping notebooks
- cleaned datasets for wrangling and EDA
- SQL, Folium, and Dash analyses
- classification-model notebook and results

All artefacts are organised in the project repository.



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Final PDF Presentation
Jack • July 2026

GitHub repository: <https://github.com/darizek-art/ibm-data-science-capstone>



Executive Summary

A data-driven assessment of Falcon 9 first-stage recovery.

Historic data show a 66.7% landing-success rate in the predictive dataset.

Three classifiers tied for the highest test accuracy: 83.3%.

Introduction

Can Falcon 9 first-stage landing success be predicted from launch characteristics?

Successful recovery matters because reused first stages can reduce launch cost.

Methodology

- Collected launch, payload, booster, and launch-site data from SpaceX sources and Wikipedia tables.
- Cleaned records, created the binary landing-success label, and prepared model features.
- Used visual and SQL analysis to identify relationships among payload, orbit, site, and outcomes.
- Built and evaluated classification models using a held-out test set.

EDA with Data Visualization

Scatter plots compared payload, flight number, launch site, and orbit type.

- Bar charts measured orbit-level landing success, while a line chart showed the yearly trend.
- These views exposed important feature relationships before modelling.

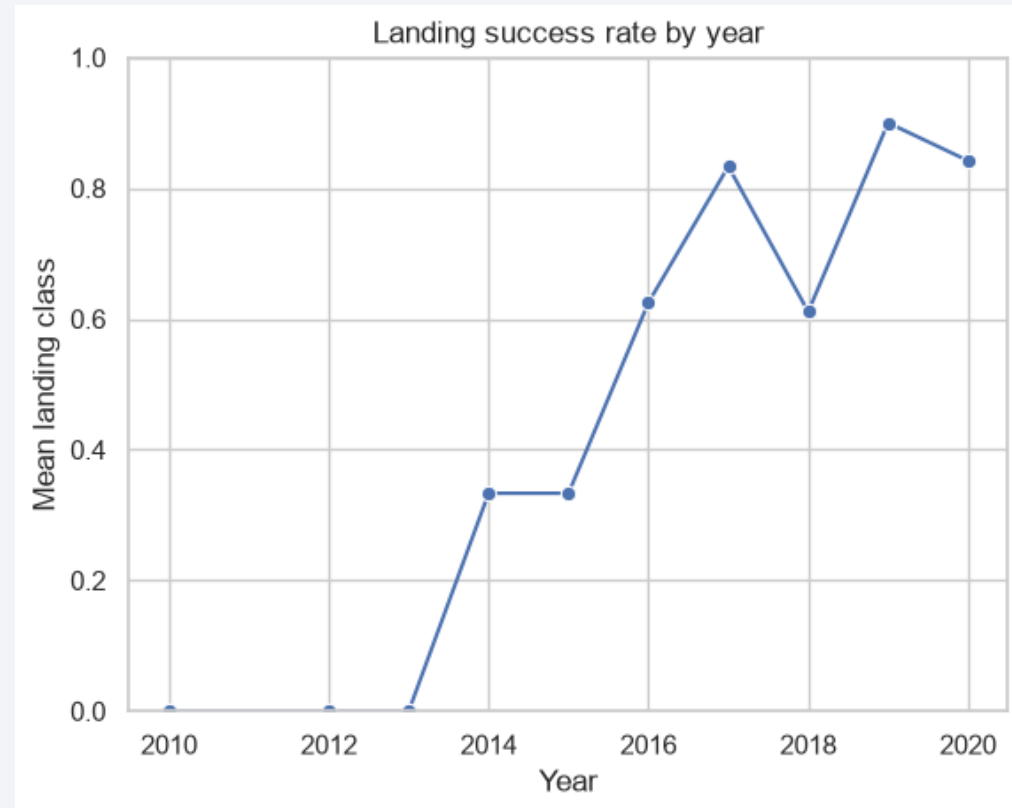
Predictive Analysis (Classification)

- Standardised features and split the data into training and test sets.
- Evaluated Logistic Regression, SVM, Decision Tree, and KNN classifiers.
- Compared test accuracy and confusion matrices to select the strongest result.

Launch Success Yearly Trend

The yearly average landing-success rate trends upward as recovery operations mature.

The pattern supports including year and flight number as predictive features.



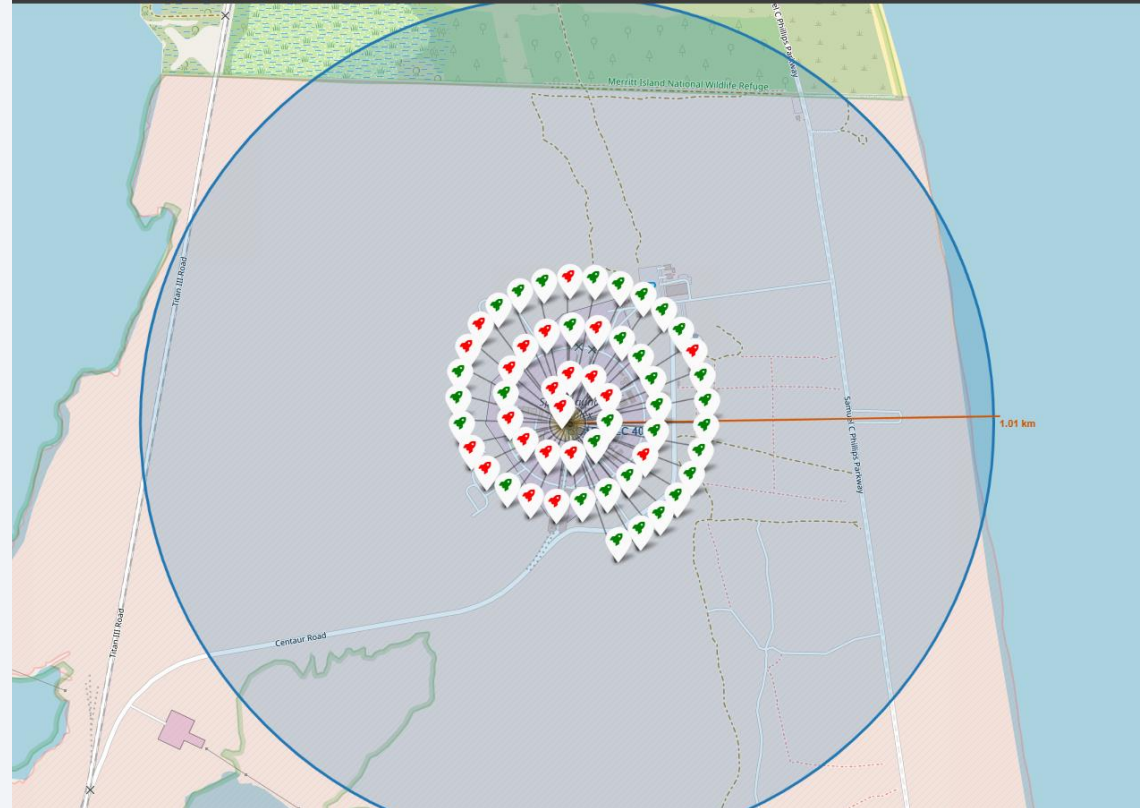
EDA with SQL Queries and Results

SQL query filter: successful drone-ship outcomes with payload mass between 4,000 and 6,000 kg.

Result: F9 FT B1026 (4,600 kg), B1022 (4,696 kg), B1031.2 (5,200 kg), and B1021.2 (5,300 kg).

Landing Outcomes by Launch Site

Green = success; red = failure.



Dash: Launch-Site Success Comparison

CCAFS SLC 40: 95% successful, 5% failed.

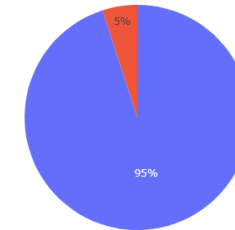
SpaceX Launch Records Dashboard

CCAFS SLC 40

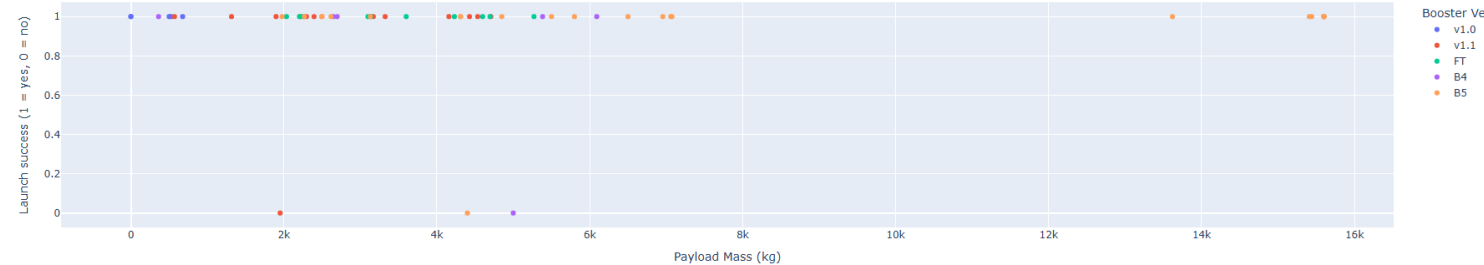
Payload range (kg)

0

Launch Outcomes for CCAFS SLC 40



Payload Mass vs. Launch Success — CCAFS SLC 40

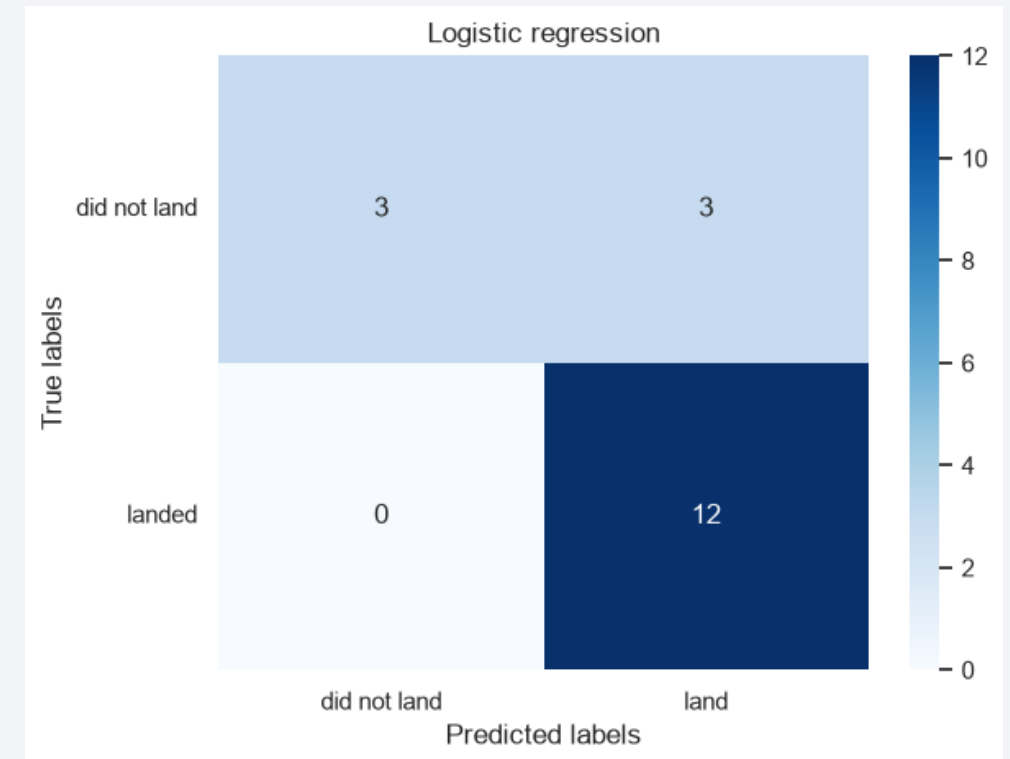


Predictive Analysis Results

Test accuracy by model:

- Logistic Regression — 83.3%
- Support Vector Machine — 83.3%
- Decision Tree — 77.8%
- K-Nearest Neighbours — 83.3%

Three models tied for the highest observed test accuracy.

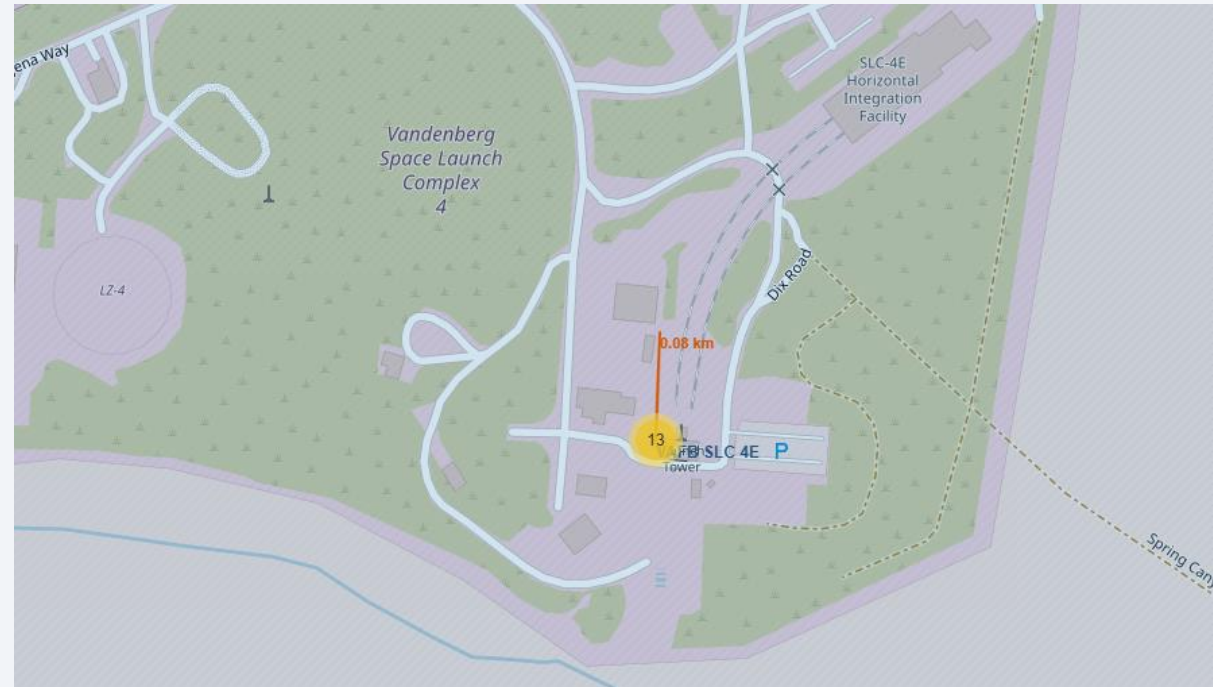


Conclusions

- Falcon 9 landing success rose as operations matured.
- Launch site, orbit, payload, and flight number are informative features.
- Historic landing data supported useful classification performance.
- The 83.3% top test score should be interpreted cautiously because the dataset is small and the best models were tied.

Creativity: Folium Proximity Analysis

Measured coast distance: 0.08 km.



Innovative Insights

Key results

- Landing success in the modelling dataset: 66.7%
- EDA showed outcome patterns by orbit, site, payload, and time
 - Interactive tools supported geographic and launch-site exploration
 - Best test accuracy: 83.3%